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IT3930E - 727528 - Project II

Federated learning on skin classification

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Contents

LIST OF ABBREVIATIONS

EHR Electronic Health Records

FedAvg Federated Averaging

ICU Intensive Care Unit

IID Independent and Identically Distributed

VGG Visual Geometry Group

Chapter 1

Introduction

The healthcare system has witnessed significant advancements in recent years, particularly in the integration of AI applications for the purposes of diagnosing, treating, and managing various medical conditions. In this dynamic landscape, researchers, clinicians, and technology developers are collaborating more than ever to unlock AI's full potential and overcome the barriers that stand in the way of its widespread adoption. As AI algorithms continuously evolve and mature, the future of healthcare holds immense promise, fostering a new era of precision medicine, personalized treatments, and improved healthcare accessibility for all.

Despite the incredible potential, the integration of AI in healthcare is not without its challenges. Concerns related to data privacy, security, and ethical considerations require careful navigation to ensure patient information remains protected while reaping the benefits of AI-driven advancements. This creates a big barrier for developing effective analytical approaches that are generalizable, which need diverse, "big data.". Therefore, there is a need for a comprehensive solution that allows for the use of data to train a model while ensuring the security and privacy of the data. This solution is called federated learning, which is a mechanism that enables the training of a shared global model with a central server while keeping all the sensitive data in local institutions where they originate. This approach holds great potential in connecting fragmented healthcare data sources while preserving privacy.

Several tasks have been explored in the field of federated learning in healthcare. These tasks include patient similarity learning, patient representation learning, phenotyping, and predictive modeling. For instance, Lee et al. [1] developed a privacy-preserving platform that allows for patient similarity learning across different institutions without sharing individual patient data. Kim et al. [2] used tensor factorization models to extract meaningful phenotypes from large-scale electronic health records in a federated learning setting. Liu et al. [3] conducted patient representation learning and obesity comorbidity phenotyping in a federated manner, achieving positive results. Vepakomma et al. [4] employed a distributed deep learning method called SplitNN to enable health entities to collaboratively train models without sharing sensitive data. Silva et al. [5] used federated learning to investigate brain structural relationships across diseases and clinical cohorts. Huang et al. [6] addressed the issue of non-IID ICU patient data by clustering patients into clinically meaningful communities and training separate models for each community. Furthermore, federated learning has enabled predictive modeling based on diverse data sources, offering clinicians additional insights into treatment risks and benefits. Brisimi et al. [7] predicted future hospitalizations for patients with heart-related diseases by leveraging EHR data from various sources in a federated learning environment. Boughorbel et al. [8] proposed a federated uncertainty-aware learning algorithm for the prediction of preterm birth using distributed EHR data. Pfohl et al. [9] focused on predicting prolonged length of stay and in-hospital mortality across multiple hospitals. Sharma et al. [10] tested a privacy-preserving framework for in-hospital mortality prediction in the ICU, demonstrating comparable performance to traditional centralized learning methods within the federated learning framework.

The objective of this project is to experiment with federated training on healthcare data and evaluate its performance on the task of classifying skin images. Beside maximizing classification metrics (accuracy, precision,..), we will also examine the impact of non-IID (non-independent-and-identically-distributed) data on federated learning strategy.

The remainder of the report is structure as follows. Chapter 2 reviews the background required to follow our work, including the model architecture and the federated schema. Chapter 3 describes the data that we used for experiments, its statistic as well as the way of splitting. In Chapter 4, a comprehensive analysis is provided regarding the methodology for training both centralized and federated strategy settings. Additionally, the chapter delves into the results obtained from the conducted experiments. Finally, we conclude our work in chapter 5.

Chapter 2

Background

2.1 VGG16

The VGG model, also known as VGGNet, supports 16 layers and is commonly referred to as VGG16. It is a convolutional neural network model proposed by A. Zisserman and K. Simonyan [11]. The number 16 in the name VGG indicates that it has a deep neural network with a total of approximately 138 million parameters, which is quite large even by modern standards. However, what makes VGG16 more attractive is its simple architecture (Fig 2.1). By examining the architecture, we can see that it follows a uniform pattern, consisting of several convolution layers followed by a pooling layer that reduces the height and width. The number of filters available to use starts at around 64 and can be doubled to 128, then to 256 in subsequent layers. In the final layers, 512 filters can be utilized.

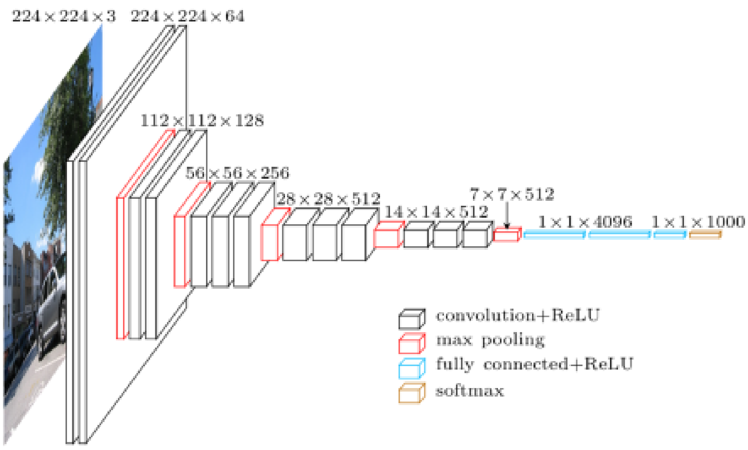


Figure 2.1: VGG Architecture [11]

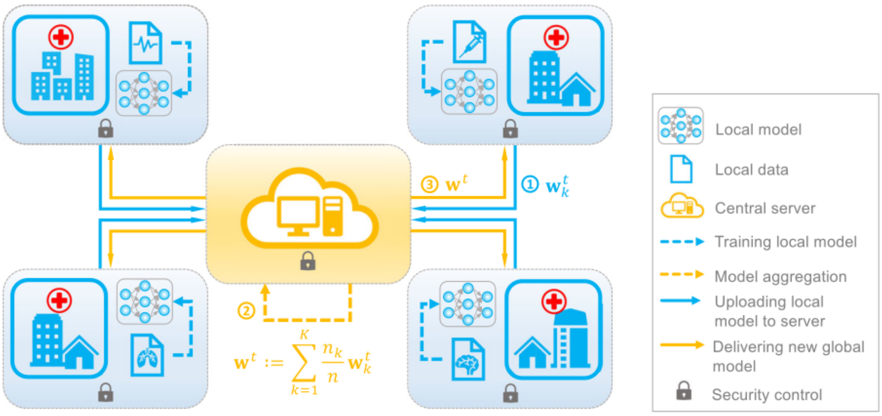


Figure 2.2: Federated learning schema [12]

2.2 Federated learning and FedAvg

Federated learning [12] is the task of training a shared global model with a central server using decentralized data that is spread across numerous clients (Fig. 2.2). Each client, such as a mobile phone or a clinical institution data warehouse, has its own data distribution represented by D_k, and the number of available samples from each client is denoted by n_k. The total sample size is given by $n = \sum_{k=1}^{K} n_k$. The objective of federated learning is to solve an empirical risk minimization problem in the form of:

$$\min_{\mathbf{w} \in \mathbb{R}^d} F(\mathbf{w}) := \sum_{k=1}^{K} \frac{n_k}{n} F_k(w) \quad \text{where} \quad F_k(\mathbf{w}) := \frac{1}{n_k} \sum_{x_i \in \mathcal{D}_k} f_i(\mathbf{w}) \tag{2.1}$$

The model parameter that needs to be learned is represented by $\mathbf{w}$. The function f_i is determined based on a loss function that takes into account a pair of input-output data, namely $\{\mathbf{x}_i, y_i\}$.

To solve 2.1, the intuitive idea is FedAvg [13], which is to average the model parameters across all the clients to create a global model. The algorithm proceeds through several rounds of communication between the clients and the central server. The equation for model aggregation (averaging) in FedAvg can be represented as follows:

$$w_{t+1} = \sum_{k \in S_t} \frac{n_k}{\sum_{k \in S_t} n_k} w_{t+1}^k \tag{2.2}$$

where S_t is random set of m clients selected at round t, w_{t+1}^k is updated parameter from client k.

Chapter 3

Data

3.1 Data format and exploration

For experimental convenient, we chose an open source data set about skin image in healthcare, Fitzpatrick17k [14]. As the name suggest, the Fitzpatrick17k contains 16,577 clinical images with skin condition labels and skin type labels based on the Fitzpatrick scoring system [15].

Upon further examination, we have discovered that the dataset exhibits a significantly imbalanced distribution of classes. Specifically, the non-neoplastic class predominates, as indicated by the data distribution presented in Figure 3.1. Since the performance

3.2 Training and validation set

For evaluating the model, the data is split for multiple parts for training, validating, testing. One factor of taking into consideration while splitting is the distribution of skin colors. Due to the fact that skin color has a significant impact on performance, to ensure equity of the evaluation, the distribution of this feature should be consistent across all datasets. Therefore, we used stratified sampling when dividing the data set. The different of stratified split and non-stratified split on centralize training is illustrated in Figure 3.1 and 3.2

In order to train in federated learning, we begin by dividing the dataset into a training set and a test set, with a ratio of 9:1. Next, we further divide the training set into multiple parts, which equal to the number of clients required for simulation. To assess how non-IID data affects the performance of the FedAvg algorithm, we employ two different approaches: non-IID splitting and IID splitting. In the case of non-IID splitting, each client is assigned data sets with the lowest skin impurity level, i.e. each client data will consist of people with 1 to 2 different skin colors. At each client, we continue to split the data, allocating 10% for validation purposes and the remaining portion for training.

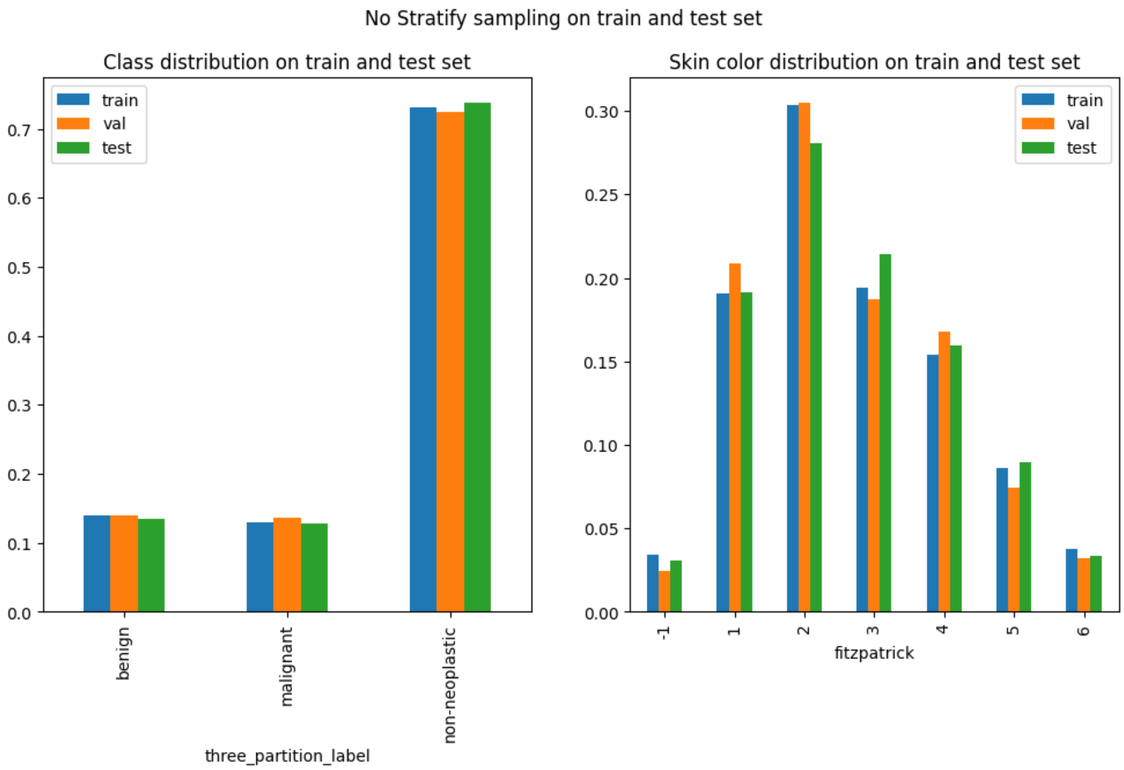


Figure 3.1: Class distribution and feature distribution of non-stratified data split

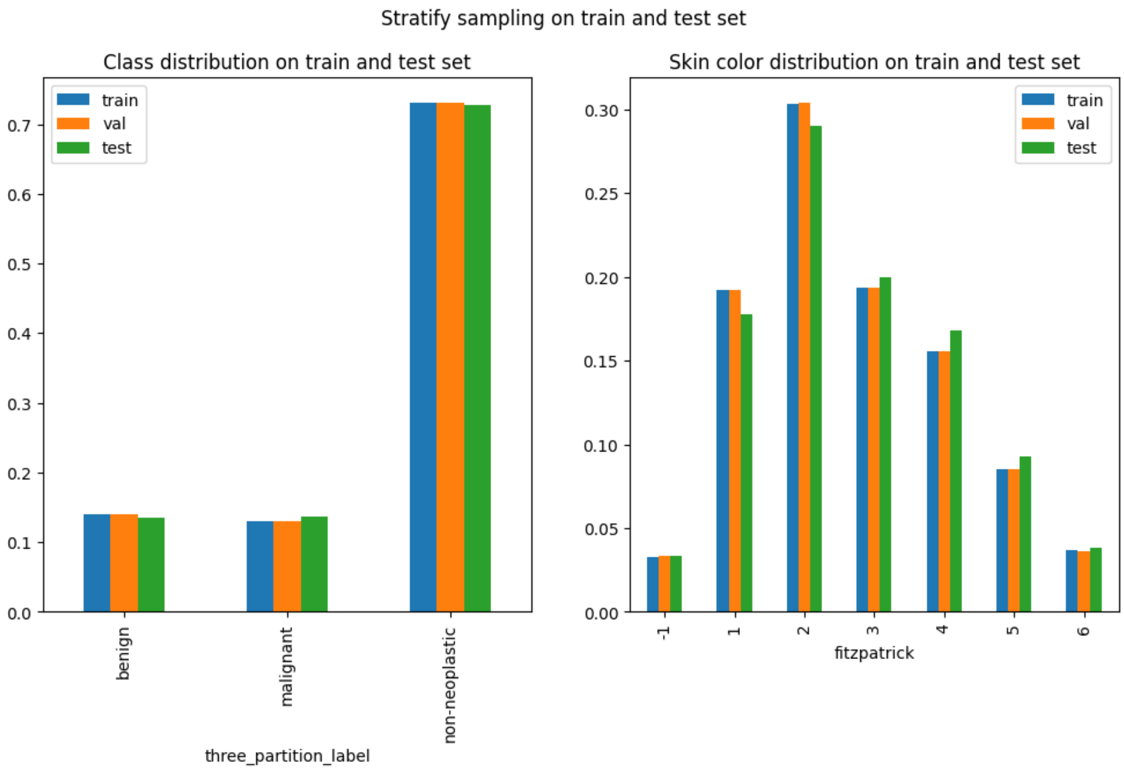


Figure 3.2: Class distribution and feature distribution of stratified data split

Chapter 4

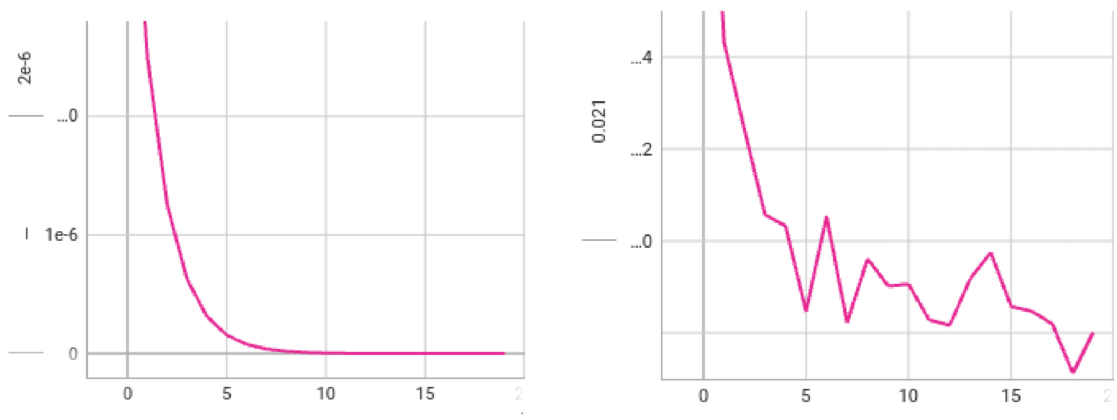
Experiments

4.1 Centralized training schema

In order to effectively training VGG16, we initiate a high learning rate of 10^{-5} (obtained through experiments) to speed up training, and reduce it exponentially by using exponential learning rate scheduler with $\gamma = 0.5$ (4.1a). To minimize the cross entropy loss, the Adam [16] optimizer is used with $(\beta_1, \beta_2) = (0.9, 0.999)$ and zero weight decay. The VGG16 is trained with batch size 32 and reach optimal point on training set after 20 epochs (4.1b).

4.2 Federated training schema

The federated learning schema requires a central server and multiple clients. In our simulation, we create a virtual network of one server and 10 clients, in which each client hold their own data created by the data splitting method presented in section 3.2. The server can request for any client to send their parameters, assume all



(a) Learning rate (b) Loss

Figure 4.1: Centralized training process

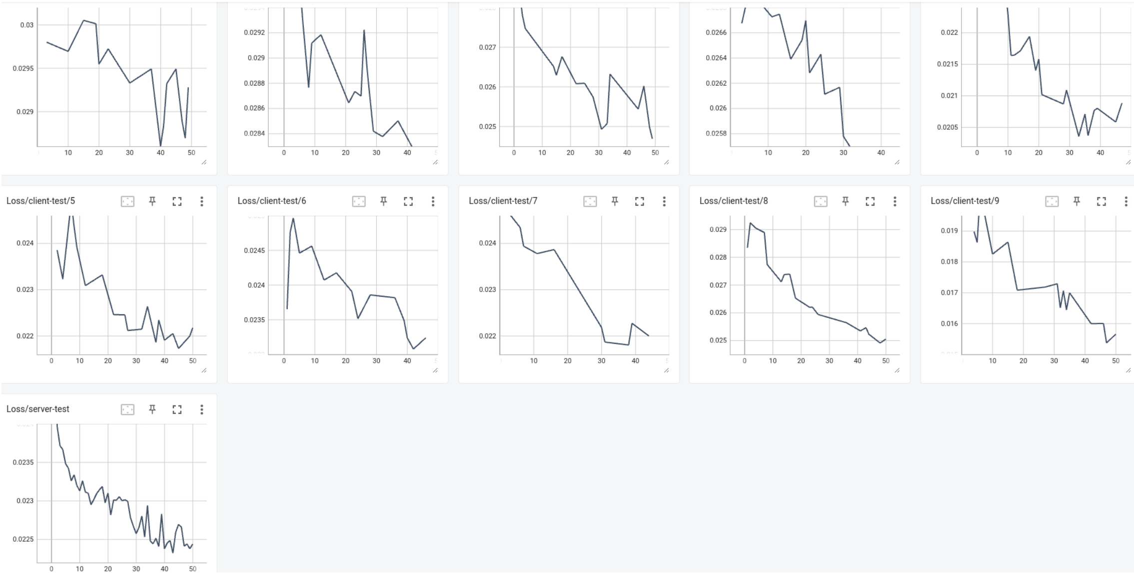


Figure 4.2: Federated training loss on server and client model

communication channels to the client remain stable throughout the training process.

We train the model under the fraction fit of 30%, which means each round the server just required 30% active client in the number of total clients to send their parameters, and average those parameters using FedAvg schema to update the central model. To simplify the training process, we change the dynamic learning rate scheduler to a constant learning rate, smaller 10 times compared to the initial learning rate of centralized learning, which is 10^{-6}. In each round, the client is trained for 2 epoch, except the first round trained for just 1 epoch. The loss and optimizer are remain the same from centralized learning with the same hyper parameters. With these setup, the model converge after 50 round, as been seen on the monitor dashboard in Fig 4.2. To estimate the effect of non-IID data to the performance of FedAvg, we apply the non-IID splitting and train it independently with the IID splitting.

4.3 Evaluating schema

We use 2 common metrics for evaluating our classification model, which is accuracy and precision. The accuracy and the precision is computed as:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \tag{4.1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.2}$$

in which we focus on the precision of the most severe class - malignant as our most important metric.

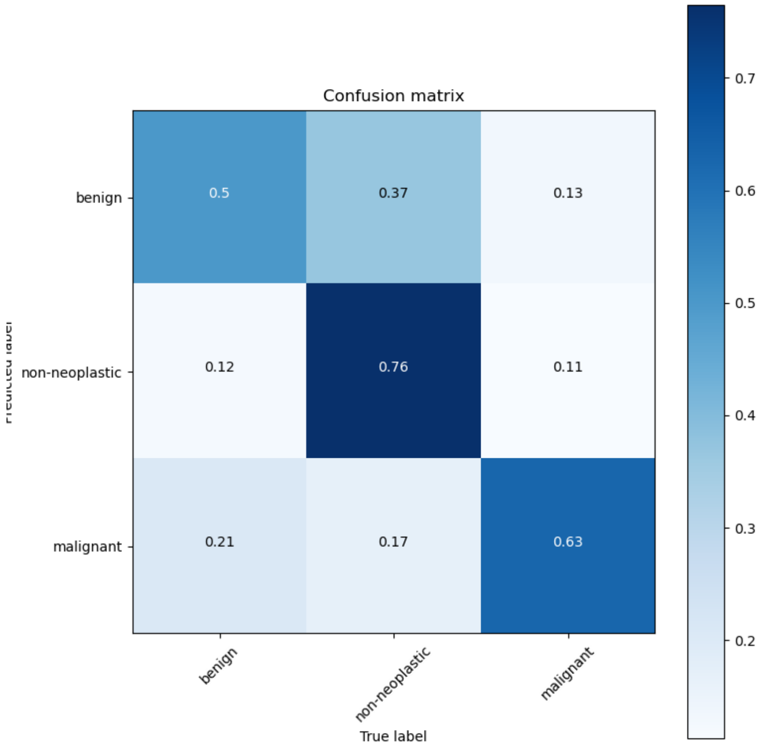


Figure 4.3: Confusion matrix on test data after centralized training

In order to monitor the progress of clients in federated learning across multiple rounds, the server will choose a random 30% of total number of clients and ask them to assess their model at the end of each round. Once the server receives the metrics from all the clients, it will combine them into a single comprehensive metric for each client. Additionally, the server will evaluate its own central model using its own test data and compare its performance with that of the clients.

4.4 Experimental result

4.4.1 Result on centralized training

The outcomes obtained through centralization are somewhat unsatisfactory. The test set's accuracy is approximately 75%, and the precision for each class is shown in Figure 4.1. Notably, the crucial metric for malignancy is just around 63%. However, this is an acceptable result considering the complexity of the VGG16 model compared to the complexity of the data that the model has to handle. The objective of this project is to explore the concept of federated learning, and these results will serve as the baseline for the method. An effective strategy for federated learning should aim to closely approach this centralized outcome while ensuring fairness among clients.

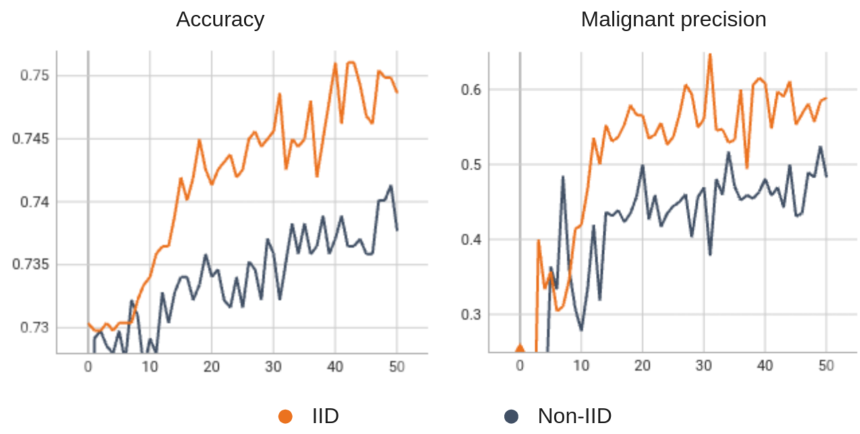


Figure 4.4: Federated learning test result on server with IID versus non-IID data

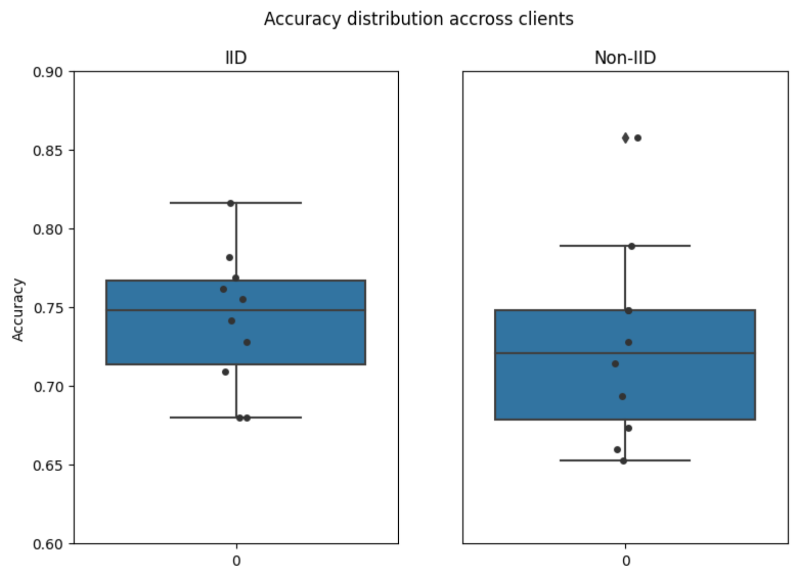


Figure 4.5: Accuracy distribution across client of IID versus non-IID data

4.4.2 Result on federated training

After 50 round of training, we see a signification impact of non-IID data on the performance of FedAvg. On one hand, when examining the performance of the central model test on the server that was trained on non-IID data (4.4), there is a noticeable decline of approximately 2% in accuracy and a more substantial decrease of around 17% in malignant precision. On the other hand, the fairness of clients is also decrease. As illustrated in Fig 4.5, the accuracy of clients on non-IID data are more dispersed, with a larger variance compare to IID. The mean accuracy is also witnessed the same pattern with the accuracy tested on server.

Chapter 5

Conclusion and future work

To sum up, our project revolves around an experiment involving federated training where we utilize the VGG16 model to classify skin images. Apart from focusing on achieving high classification metrics such as accuracy and precision, we also investigate how the presence of non-IID data affects the effectiveness of the federated learning strategy.

In our future endeavors, we aim to enhance our work in several key areas. Firstly, we plan to optimize more intricate architectures like ConvNext [17] and EfficientNetV2 [18] by employing larger batch sizes and increased RAM capacity. This optimization will allow us to achieve even higher levels of accuracy and precision in our models. Additionally, we recognize the challenge posed by non-IID (non-identically distributed) data in federated learning. To address this issue, we will explore alternative federated learning strategies that can effectively handle non-IID data. Lastly, we intend to experiment with various data augmentation techniques at the client level to further enhance the quality and diversity of the data used in our models. These future endeavors will undoubtedly contribute to the advancement of our research and the performance of our machine learning solutions.

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